

1. Civil Works

1.1 Site Preparation

- Surveying
- Site Clearance
- Demolition
- Excavation
- Dewatering
- Earthworks
- Backfilling
- Soil Improvement

1.2 Foundations

- Piling Works
- Bored Piles
- Driven Piles
- Pile Caps
- Raft Foundations
- Isolated Footings
- Combined Footings
- Ground Beams

1.3 Substructure

- Basement Structures
- Retaining Walls
- Waterproofing
- Underground Pits
- Underground Tanks

1.4 Superstructure (Concrete)

- Columns
- Shear Walls
- Core Walls
- Beams
- Slabs

- Transfer Structures
- Staircases
- Ramps

1.5 Architectural Civil Works

- Blockwork
 - Masonry
 - Plastering
 - Screeding
 - Raised Floors
 - Waterproofing
 - Roofing Systems
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2. Structural Steel Works

2.1 Steel Fabrication

- Shop Drawings
- Material Procurement
- Fabrication
- Surface Treatment
- Painting
- Fireproofing Preparation

2.2 Steel Erection

- Steel Columns
- Steel Beams
- Trusses
- Bracings
- Platforms
- Catwalks
- Roof Structures

2.3 Connection Works

- Bolted Connections

- Welded Connections
- Base Plates
- Anchor Bolts

2.4 Specialized Steel Structures

- Space Frames
 - Canopies
 - Airport Roof Structures
 - Metro Station Steel Structures
 - Pipe Racks
 - Equipment Support Structures
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3. MEP

3.1 Mechanical Systems

HVAC

- Central Chiller Plant
- Cooling Towers
- Chilled Water Network
- Condenser Water Network
- AHUs
- FCUs
- VRF / VRV Systems
- Precision Air Conditioning (Data Centers)
- Ventilation Systems
- Kitchen Ventilation
- Car Park Ventilation
- Smoke Management Systems
- Pressurization Systems

Fire Fighting

- Fire Water Tanks
- Fire Pumps
- Sprinkler Systems
- Hydrant Systems

- Hose Reel Systems
- Foam Systems
- Clean Agent Systems (FM200 / Novec)
- Water Mist Systems

Fuel & Gas

- Diesel Fuel Systems
- LPG Systems
- Natural Gas Systems

Specialized Mechanical Systems

- Medical Gas Systems (Hospitals)
 - Pneumatic Tube Systems (Hospitals)
 - Baggage Handling Systems (Airports)
 - Platform Screen Door HVAC Integration (Metro)
 - Data Center Cooling Systems
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3.2 Electrical Systems

HV Systems

- Utility Interface
- GIS / AIS Substations
- HV Switchgear
- HV Cabling

MV Systems

- Transformers
- Ring Main Units (RMU)
- MV Switchgear

LV Systems

- MDBs
- SMDBs
- Final Distribution Boards
- Power Cabling

Emergency Power

- Diesel Generators
- UPS Systems
- Battery Systems
- Static Transfer Switches

Lighting

- Internal Lighting
- External Lighting
- Emergency Lighting
- Architectural Lighting
- Airfield Lighting (Airports)

Earthing & Lightning Protection

- Grounding Network
- Lightning Protection Systems

Renewable Energy

- Solar PV Systems
 - Energy Storage Systems
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3.3 ELV / ICT Systems

Structured Cabling

- Copper Network
- Fiber Optic Backbone
- Data Center Cabling

Security Systems

- CCTV
- Access Control
- Intrusion Detection
- Perimeter Security

Communication Systems

- Intercom
- Public Address

- IPTV
- SMATV

IT Infrastructure

- Server Rooms
- Data Centers
- Network Equipment Rooms

Building Automation

- Building Management System (BMS)
- Energy Management System (EMS)

Life Safety Systems

- Fire Alarm System
- Voice Evacuation System
- Emergency Communication System

Specialized ELV

- Passenger Information System (Metro)
 - Flight Information Display System (Airports)
 - Nurse Call System (Hospitals)
 - Queue Management Systems
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3.4 Plumbing Systems

Water Supply

- Domestic Cold Water
- Domestic Hot Water
- Booster Pump Systems
- Water Storage Tanks

Drainage

- Soil & Waste Systems
- Vent Systems
- Storm Water Systems
- Rainwater Systems

Sanitary Systems

- Fixtures
- Water Heating Systems
- Grease Waste Systems

Irrigation

- Landscape Irrigation
- Treated Water Irrigation

Water Treatment

- RO Plants
- Water Softening
- Filtration Systems

Wastewater Treatment

- Sewage Treatment Plants (STP)
 - Grey Water Treatment Systems
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4. Building Envelope

4.1 Facade Systems

- Curtain Walls
- Unitized Facade
- Stick Facade
- Glazing Systems

4.2 Cladding Systems

- Aluminum Cladding
- Stone Cladding
- GRC Cladding
- Metal Panels

4.3 Roofing Systems

- Metal Roofing
 - Waterproof Membranes
 - Insulation Systems
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5. Infrastructure Works

5.1 Roads & Pavements

- Road Construction
- Asphalt Paving
- Concrete Paving
- Kerbs
- Interlock Paving

5.2 Utilities

- Potable Water Networks
- Irrigation Networks
- Sewer Networks
- Stormwater Networks
- Utility Corridors

5.3 External Works

- Landscaping
- Hardscape
- Fencing
- Gates
- Site Furniture

5.4 Underground Structures

- Utility Tunnels
- Culverts
- Service Ducts
- Chambers
- Manholes

6. Temporary Works

6.1 Excavation Support

- Shoring
- Sheet Piling
- Soldier Piles
- Diaphragm Walls

6.2 Construction Support

- Formwork
 - Scaffolding
 - Falsework
 - Temporary Platforms
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7. Finishing Works (Civil Scope)

7.1 Internal Finishes

- Floor Finishes
- Wall Finishes
- Ceiling Systems
- Painting

7.2 External Finishes

- External Paving
 - External Architectural Features
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8. Specialized Systems by Project Type

Hospitals

- Medical Gas Systems
- Nurse Call Systems
- Isolation Room Pressure Control
- Pneumatic Tube Systems
- Specialized HVAC for Operating Theaters

Airports

- Baggage Handling Systems
- Airfield Lighting
- Flight Information Systems
- Fuel Hydrant Systems
- Large-Scale Smoke Control Systems

Metro Stations

- Tunnel Ventilation Systems
- Traction Power Interfaces
- Platform Screen Door Systems
- Passenger Information Systems
- Emergency Ventilation Systems

Data Centers

- Precision Cooling (CRAC/CRAH)
- UPS Systems
- Busway Systems
- Clean Agent Fire Suppression
- DCIM (Data Center Infrastructure Management)